

5.3.18 Delay block (DLYB) characteristics

Unless otherwise specified, the parameters given in Table 55 for the delay block are derived from tests performed under the ambient temperature and V_{DD} supply voltage summarized in Table 1.

Table 55. DLYB characteristics

Evaluated by characterization, not tested in production unless otherwise specified.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
t_{init}	Initial delay	-	950	1400	1750	ps
t_{Δ}	Unit delay	-	40	52	64	ps

5.3.19 12-bit analog-to-digital converter ADC characteristics

Unless otherwise specified, the parameters given in Table 56 are values derived from tests performed under ambient temperature, f_{HCLK} frequency, and V_{DDA} supply voltage conditions summarized in Table 16.

Note: It is recommended to perform a calibration after each power-up.

Table 56. 12-bit ADC characteristics

Specified by design - Not tested in production.

Symbol	Parameter	Conditions						Min	Typ	Max	Unit
V _{DDA}	Analog power supply for ADC ON	-						2.70	-	3.6	V
V _{REF+}	Positive reference voltage	-						2.5	-	V _{DDA}	
V _{REF-}	Negative reference voltage	-						V _{SSA}			
f _{ADC}	ADC clock frequency	2.7V ≤ V _{DDA} ≤ 3.6 V						8	-	36	MHz
f _S with R _{AIN} =47 Ω and C _{PCB} = 22 pF	Sampling rate for slow channels	Resolution = 12 bits	All modes	2.7 V ≤ V _{DDA} ≤ 3.6 V	-40°C ≤ T _J ≤ 140°C	f _{ADC} = 36 MHz	SMP = 2.5	-	2.25	-	MSPS
		Resolution = 10 bits						-	2.57	-	
		Resolution = 8 bits						-	3	-	
		Resolution = 6 bits						-	4.5	-	
t _{TRIG}	External trigger period	Resolution = 12 bits						16	-	-	1/f _{ADC}
V _{AIN}	Conversion voltage range	-						0	-	V _{REF+}	V
R _{AIN} ⁽¹⁾	External input impedance	Resolution = 12 bits, T _J = 140°C						-	-	110	Ω
		Resolution = 12 bits, T _J = 125°C						-	-	610	
		Resolution = 10 bits, T _J = 140°C						-	-	2305	
		Resolution = 10 bits, T _J = 125°C						-	-	4290	
		Resolution = 8 bits, T _J = 140°C						-	-	11110	
		Resolution = 8 bits, T _J = 125°C						-	-	15860	
		Resolution = 6 bits, T _J = 140°C						-	-	46890	